Stat 184 Project- NFL Statistics

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Project Overview

This is a quarto document where we input our findings about offense statistics in the NFL and comparing them to the win percentage. There will be three major sections that have visualizations of our findings and the reasoning behind them.

Background Information

We took data from SCORE Sports Data Repository and took two sets of data for our project. The first was a data set that ranged from 2018-2023 that had all regular season games between all teams and had the outcome of when the team was either away or home. The second was a data set that ranged from 1999-2022 that had a variety of statistics for each team. This included all offense data, defense data, wins, losses, etc.

References

For this data, we got our data from SCORE Data Sports Repository, which both of these files were included within our own repository. https://data.scorenetwork.org/

Also, for the descriptions for the pie chart data was obtained from ChatGPT. The prompt was "Can you generate alt text and narrative text for this pie chart for nfl data from 2018-2022". This was generated at 9:45pm 5/6/2026.

Average Offensive Performance by Game Outcome
NFL Seasons 2018–2022

Game Result	Avg Points	Avg Passing Yards	Avg Rushing Yards
Losses	21.98	6.14	4.48
Wins	24.37	6.48	4.57

Passing Performance by Game Outcome
NFL Seasons 2018-2022

Game Result	Avg Passing Yards	Avg Points
Losses	6.14	21.98
Wins	6.48	24.37

Running Performance by Game Outcome
NFL Seasons 2018-2022

Game Result	Avg Running Yards	Avg Points
Losses	4.48	21.98
Wins	4.57	24.37

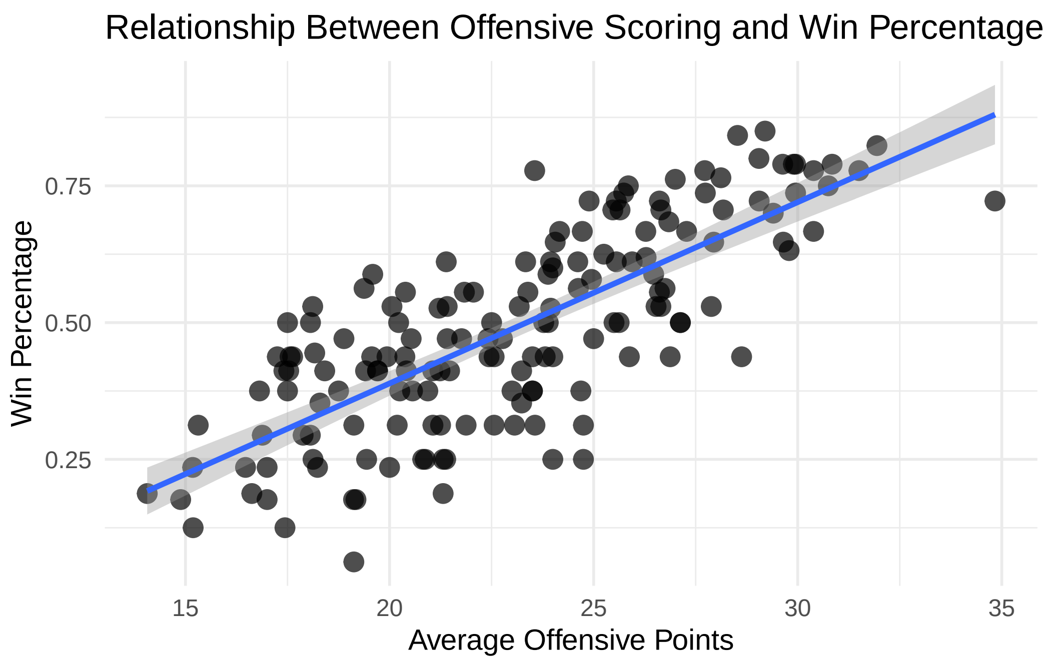
Frequency Table

Each statistic shown in the table is specifically about the offense. The row is labeled as either Win or Loss. This is to show on average how the offense for each team performed and then the outcome of the game. From the table, it is evident that when the numbers for how the offense performs are higher, the team wins the game.

Alt Text: Table represents all teams' offensive stats when they win or lose their game.

This table shows different statistics based on the offense of each team in the NFL from the years 2018-2022. Yards gained is how many yards the offense moved forward on each play, getting closer to the end zone. There are two different kinds of ways to gain yards; there's pass and run. The quarterback has the option to either throw the ball or give it to another player to run it through the defense. Average Air Yards is how far the football travels when the quarterback throws it to his teammate. Average YAC is Yards After Catch; this means it's finding the average yards the receiver ran after the ball was caught. EPA is Expected Points Added; this means for either a pass play or run how many points they are expected to earn from that play. WPA is Win Probability Added; which in this case only takes into consideration how the offence performance determines the chance of them winning the game overall. Average points are the average amount of points earned overall. Knowing all of this, when looking at the table it is evident that the stats are higher for the offense when the outcome of the game is a win. From this, we can determine how the offense plays are positively associated with a win.

Average Points Per Game vs Win Percentage



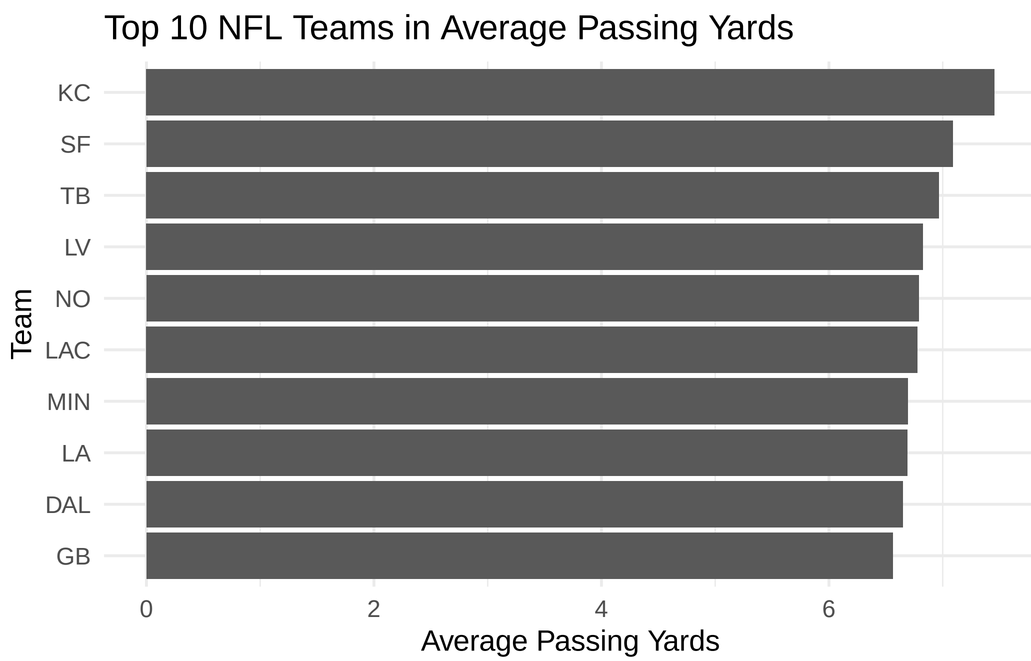
The image is a scatter plot graph illustrating the relationship between the average PPG and win ratio for the offense from each NFL team from 2018-2022. The horizontal axis represents the average offensive points and has a range of 7 to 35 points. The vertical axis represents the win percentage, which ranges from 0-1. Each glyph shown on the graph represents an NFL team (32 teams total). The line in the middle of the graph is the overall average from 2018-2022, so if a team falls below the line, they are below the average. If they are above the line, then they are above average. From this it is indicated that the more average offensive points there are, the higher the win percentage is.

Alt Text: Scatter plot showing a positive relationship between average PPG (Points Per Game) and win ratio.

In the figure, we can examine the potential relationship between average PPG and win percentage for each NFL team over the course of 2018-2022. There is a total of 32 teams that are being represented from each glyph on the graph. Each team is being measured by the same scale, which is why we can draw a conclusion on the relationship between PPG and win percentage. We can conclude that the higher the PPG is, the higher the win percentage is. However, there appear to be some discrepancies where this isn't the case. On average though, this appears to be the case.

Average Offensive Yards (Pass vs Run)

Passing Yards

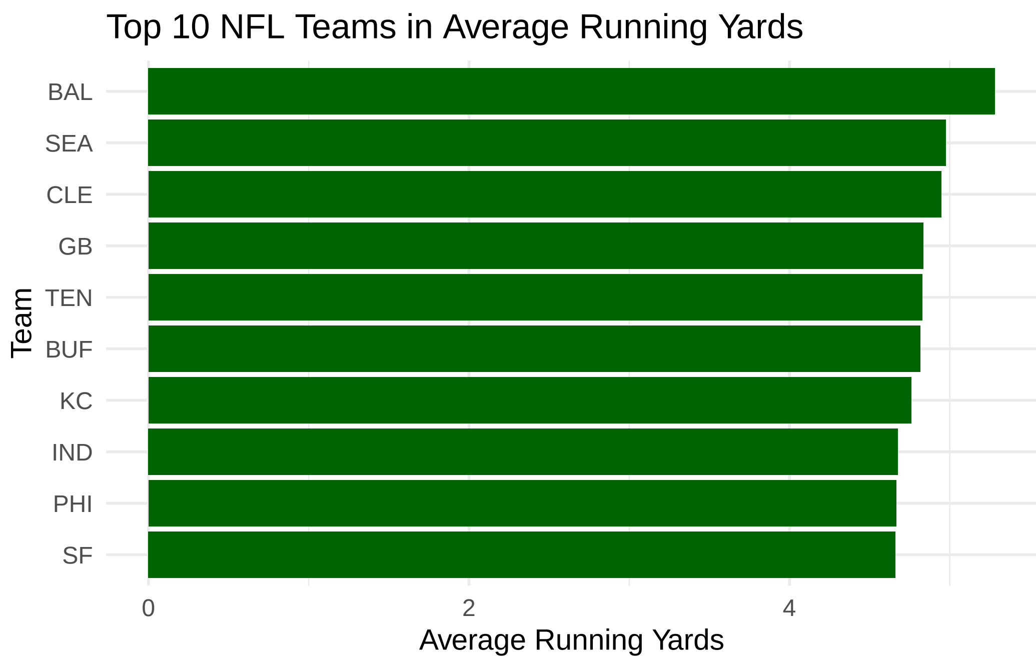


The image is a horizontal bar graph illustrating the top 10 NFL teams with the highest average offensive passing yards from 2018–2022. The horizontal axis represents the average passing yards gained, while the vertical axis lists the NFL teams. Each bar represents one team and its overall passing performance during the selected seasons. Teams with longer bars had higher average passing yard totals. From the graph, it is evident that some teams were significantly more effective in the passing game than others.

Alt text: Horizontal bar chart displaying the top 10 NFL teams in average passing yards from 2018–2022.

In the figure, we can compare the average offensive passing yards among the top passing teams in the NFL from 2018–2022. Each bar represents one NFL team and allows us to visually compare their offensive passing production. The teams near the top of the graph had the highest average passing yards, indicating stronger and more efficient passing offenses during this time period. Although the graph does not directly display win percentage, teams with higher passing yard averages often had stronger overall offensive performances. This demonstrates how passing yards can contribute to a team's offensive success.

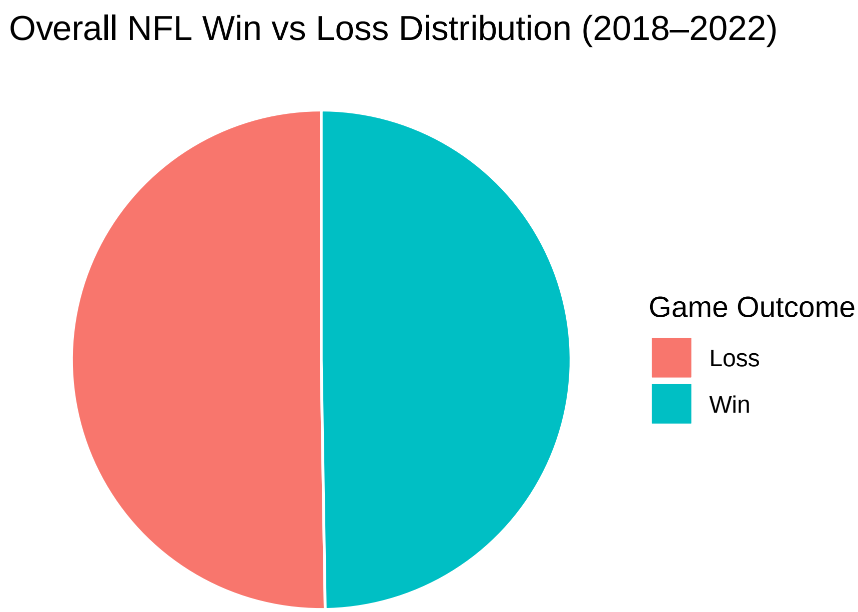
Running Yards



The image is a horizontal bar graph illustrating the top 10 NFL teams with the highest average offensive running yards from 2018–2022. The horizontal axis represents the average running yards gained, while the vertical axis lists the NFL teams. Each bar represents one team and its overall average rushing performance across the seasons included in the data. Teams with longer bars had higher average rushing yards. From the graph, it is evident that some teams relied more heavily on the running game and were more successful at gaining rushing yards than others.

Alt text: Horizontal bar chart displaying the top 10 NFL teams in average running yards from 2018–2022.

In the figure, we can compare the average offensive running yards among the top rushing teams in the NFL from 2018–2022. Each bar represents one NFL team and allows us to visually compare their rushing performance. The teams near the top of the graph had the highest average running yards, indicating stronger rushing offenses during this time period. While the graph does not directly show win percentage, it demonstrates which teams were most effective at gaining yards on the ground. This helps provide insight into how important the running game can be to an offense's overall success.



The image is a pie chart illustrating the overall distribution of NFL game outcomes (wins vs losses) from 2018–2022. The chart is divided into two colored sections representing the total number of wins and total number of losses recorded across all teams and seasons in the dataset. Each section's size corresponds to how frequently that outcome occurred. Because every game produces one win and one loss across the two teams involved, the distribution is expected to be nearly balanced overall. This visualization helps show the total structure of outcomes in the dataset before breaking down team-level performance.

Pie chart showing the overall distribution of NFL game outcomes (wins and losses) from 2018–2022, with two nearly equal sections representing each outcome.

In this figure, we can observe the overall distribution of wins and losses across all NFL games from 2018–2022. Since every game includes one winning team and one losing team, the totals for wins and losses are expected to be very similar, which is reflected in the chart. This visualization helps establish a baseline understanding of the dataset before analyzing more specific relationships between offensive performance and winning outcomes. While it does not show team-level differences, it confirms the balance of outcomes across the seasons included in the study.

Code Appendix

```
library(tidyverse)
library(readr)
library(ggplot2)
library(knitr)
library(gt)

#Loading in our datasets
```

```r
##This includes Game outcomes data and Team statistics data
gameOutcomesRaw <- read.csv("nfl_mahomes_era_games.csv")
teamStatisticsRaw <- read.csv("nfl-team-statistics.csv")

#Filtertinr the Raw data to show only years 2018 to 2022
gameOutcomesClean <- gameOutcomesRaw |>
  filter(season %in% c(2018, 2019, 2020, 2021, 2022))

#Filtertinr the Raw data to show only years 2018 to 2022 and selecting only offensive columns
#as well as specific team and season column
teamStatisticsCleaned <- teamStatisticsRaw |>
  filter(season %in% c(2018, 2019, 2020, 2021, 2022)) |>
  select(season, team, starts_with("offense_ave"))

# Convert games into team-level outcomes
teamGameOutcomes <- gameOutcomesClean |>
  select(season, home_team, away_team, home_score, away_score) |>

  # Home team rows
  transmute(
    season,
    team = home_team,
    result = if_else(home_score > away_score, "Win", "Loss")
  ) |>

  # Add away team rows
  bind_rows(
    gameOutcomesClean |>
      select(season, home_team, away_team, home_score, away_score) |>
      transmute(
        season,
        team = away_team,
        result = if_else(away_score > home_score, "Win", "Loss")
      )
  )

# Create team scoring table
teamScoring <- gameOutcomesClean |>

  # Home team points
  transmute(
    season,
    team = home_team,
    points_scored = home_score
  ) |>

  # Add away team points
```

```r
  bind_rows(
    gameOutcomesClean |>
      transmute(
        season,
        team = away_team,
        points_scored = away_score
      )
  ) |>

  # Average by team-season
  group_by(season, team) |>
  summarise(
    offense_ave_points = mean(points_scored, na.rm = TRUE)
  )

# Joining with offensive stats
teamPerformance <- teamGameOutcomes |>
  left_join(teamStatisticsCleaned, by = c("season", "team")) |>
  left_join(teamScoring, by = c("season", "team"))


# Create summary table
summaryTable <- teamPerformance |>
  group_by(result) |>
  summarise(
    Avg_Points = mean(offense_ave_points, na.rm = TRUE),
    Avg_Pass_Yards = mean(offense_ave_yards_gained_pass, na.rm = TRUE),
    Avg_Run_Yards = mean(offense_ave_yards_gained_run, na.rm = TRUE),
    .groups = "drop"
  )

passingTable <- teamPerformance |>
  group_by(result) |>
  summarise(
    Avg_Passing_Yards = mean(offense_ave_yards_gained_pass, na.rm = TRUE),
    Avg_Points = mean(offense_ave_points, na.rm = TRUE),
    .groups = "drop"
  )


runningTable <- teamPerformance |>
  group_by(result) |>
  summarise(
    Avg_Running_Yards = mean(offense_ave_yards_gained_run, na.rm = TRUE),
    Avg_Points = mean(offense_ave_points, na.rm = TRUE),
    .groups = "drop"
  )
```

```r
# Create plot
teamWinRates <- teamGameOutcomes |>
  group_by(season, team) |>
  summarise(
    wins = sum(result == "Win"),
    games = n(),
    win_pct = wins / games
  )

teamSeasonAnalysis <- teamWinRates |>
  left_join(teamStatisticsCleaned, by = c("season", "team")) |>
  left_join(teamScoring, by = c("season", "team"))
```

```r
topPassingTeams <- teamSeasonAnalysis |>
  group_by(team) |>
  summarise(
    avg_pass_yards = mean(offense_ave_yards_gained_pass, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(desc(avg_pass_yards)) |>
  slice_head(n = 10)

winLossTotals <- teamGameOutcomes |>
  count(result)
```

```r
summaryTable |>
  mutate(result = ifelse(result == "Win", "Wins", "Losses")) |>
  gt() |>
  tab_header(
    title = "Average Offensive Performance by Game Outcome",
    subtitle = "NFL Seasons 2018-2022"
  ) |>
  fmt_number(
    columns = where(is.numeric),
    decimals = 2
  ) |>
  cols_label(
    result = "Game Result",
    Avg_Points = "Avg Points",
    Avg_Pass_Yards = "Avg Passing Yards",
```

```r
    Avg_Run_Yards = "Avg Rushing Yards"
  )



passingTable |>
  mutate(result = ifelse(result == "Win", "Wins", "Losses")) |>
  gt() |>
  tab_header(
    title = "Passing Performance by Game Outcome",
    subtitle = "NFL Seasons 2018-2022"
  ) |>
  fmt_number(
    columns = where(is.numeric),
    decimals = 2
  ) |>
  cols_label(
    result = "Game Result",
    Avg_Passing_Yards = "Avg Passing Yards",
    Avg_Points = "Avg Points"
  )


runningTable |>
  mutate(result = ifelse(result == "Win", "Wins", "Losses")) |>
  gt() |>
  tab_header(
    title = "Running Performance by Game Outcome",
    subtitle = "NFL Seasons 2018-2022"
  ) |>
  fmt_number(
    columns = where(is.numeric),
    decimals = 2
  ) |>
  cols_label(
    result = "Game Result",
    Avg_Running_Yards = "Avg Running Yards",
    Avg_Points = "Avg Points"
  )
ggplot(teamSeasonAnalysis, aes(x = offense_ave_points, y = win_pct)) +
  geom_point(size = 3, alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Relationship Between Offensive Scoring and Win Percentage",
    x = "Average Offensive Points",
    y = "Win Percentage"
  ) +
```

```r
  theme_minimal()


topPassingTeams <- teamSeasonAnalysis |>
  group_by(team) |>
  summarise(
    avg_pass_yards = mean(offense_ave_yards_gained_pass, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(desc(avg_pass_yards)) |>
  slice_head(n = 10)

ggplot(topPassingTeams,
       aes(x = reorder(team, avg_pass_yards),
           y = avg_pass_yards)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Top 10 NFL Teams in Average Passing Yards",
    x = "Team",
    y = "Average Passing Yards"
  ) +
  theme_minimal()
topRunningTeams <- teamSeasonAnalysis |>
  group_by(team) |>
  summarise(
    avg_run_yards = mean(offense_ave_yards_gained_run, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(desc(avg_run_yards)) |>
  slice_head(n = 10)

ggplot(topRunningTeams,
       aes(x = reorder(team, avg_run_yards),
           y = avg_run_yards)) +
  geom_col(fill = "darkgreen") +
  coord_flip() +
  labs(
    title = "Top 10 NFL Teams in Average Running Yards",
    x = "Team",
    y = "Average Running Yards"
  ) +
  theme_minimal()

ggplot(winLossTotals,
       aes(x = "", y = n, fill = result)) +
  geom_col(width = 1, color = "white") +
```

```r
  coord_polar(theta = "y") +
  labs(
    title = "Overall NFL Win vs Loss Distribution (2018-2022)",
    fill = "Game Outcome"
  ) +
  theme_void()

library(tidyverse)
library(readr)
library(ggplot2)
library(knitr)
library(gt)

#Loading in our datasets
##This includes Game outcomes data and Team statistics data
gameOutcomesRaw <- read.csv("nfl_mahomes_era_games.csv")
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#Filtertinr the Raw data to show only years 2018 to 2022 and selecting only offensive columns
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teamStatisticsCleaned <- teamStatisticsRaw |>
  filter(season %in% c(2018, 2019, 2020, 2021, 2022)) |>
  select(season, team, starts_with("offense_ave"))

# Convert games into team-level outcomes
teamGameOutcomes <- gameOutcomesClean |>
  select(season, home_team, away_team, home_score, away_score) |>

  # Home team rows
  transmute(
    season,
    team = home_team,
    result = if_else(home_score > away_score, "Win", "Loss")
  ) |>

  # Add away team rows
  bind_rows(
    gameOutcomesClean |>
      select(season, home_team, away_team, home_score, away_score) |>
      transmute(
        season,
        team = away_team,
        result = if_else(away_score > home_score, "Win", "Loss")
```

```r
    )
  )

# Create team scoring table
teamScoring <- gameOutcomesClean |>

  # Home team points
  transmute(
    season,
    team = home_team,
    points_scored = home_score
  ) |>

  # Add away team points
  bind_rows(
    gameOutcomesClean |>
      transmute(
        season,
        team = away_team,
        points_scored = away_score
      )
  ) |>

  # Average by team-season
  group_by(season, team) |>
  summarise(
    offense_ave_points = mean(points_scored, na.rm = TRUE)
  )

# Joining with offensive stats
teamPerformance <- teamGameOutcomes |>
  left_join(teamStatisticsCleaned, by = c("season", "team")) |>
  left_join(teamScoring, by = c("season", "team"))


# Create summary table
summaryTable <- teamPerformance |>
  group_by(result) |>
  summarise(
    Avg_Points = mean(offense_ave_points, na.rm = TRUE),
    Avg_Pass_Yards = mean(offense_ave_yards_gained_pass, na.rm = TRUE),
    Avg_Run_Yards = mean(offense_ave_yards_gained_run, na.rm = TRUE),
    .groups = "drop"
  )

passingTable <- teamPerformance |>
  group_by(result) |>
```

```r
  summarise(
    Avg_Passing_Yards = mean(offense_ave_yards_gained_pass, na.rm = TRUE),
    Avg_Points = mean(offense_ave_points, na.rm = TRUE),
    .groups = "drop"
  )


runningTable <- teamPerformance |>
  group_by(result) |>
  summarise(
    Avg_Running_Yards = mean(offense_ave_yards_gained_run, na.rm = TRUE),
    Avg_Points = mean(offense_ave_points, na.rm = TRUE),
    .groups = "drop"
  )



# Create plot
teamWinRates <- teamGameOutcomes |>
  group_by(season, team) |>
  summarise(
    wins = sum(result == "Win"),
    games = n(),
    win_pct = wins / games
  )

teamSeasonAnalysis <- teamWinRates |>
  left_join(teamStatisticsCleaned, by = c("season", "team")) |>
  left_join(teamScoring, by = c("season", "team"))



topPassingTeams <- teamSeasonAnalysis |>
  group_by(team) |>
  summarise(
    avg_pass_yards = mean(offense_ave_yards_gained_pass, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(desc(avg_pass_yards)) |>
  slice_head(n = 10)

winLossTotals <- teamGameOutcomes |>
  count(result)


summaryTable |>
  mutate(result = ifelse(result == "Win", "Wins", "Losses")) |>
```

```r
  gt() |>
  tab_header(
    title = "Average Offensive Performance by Game Outcome",
    subtitle = "NFL Seasons 2018-2022"
  ) |>
  fmt_number(
    columns = where(is.numeric),
    decimals = 2
  ) |>
  cols_label(
    result = "Game Result",
    Avg_Points = "Avg Points",
    Avg_Pass_Yards = "Avg Passing Yards",
    Avg_Run_Yards = "Avg Rushing Yards"
  )
```

```r
passingTable |>
  mutate(result = ifelse(result == "Win", "Wins", "Losses")) |>
  gt() |>
  tab_header(
    title = "Passing Performance by Game Outcome",
    subtitle = "NFL Seasons 2018-2022"
  ) |>
  fmt_number(
    columns = where(is.numeric),
    decimals = 2
  ) |>
  cols_label(
    result = "Game Result",
    Avg_Passing_Yards = "Avg Passing Yards",
    Avg_Points = "Avg Points"
  )
```

```r
runningTable |>
  mutate(result = ifelse(result == "Win", "Wins", "Losses")) |>
  gt() |>
  tab_header(
    title = "Running Performance by Game Outcome",
    subtitle = "NFL Seasons 2018-2022"
  ) |>
  fmt_number(
    columns = where(is.numeric),
    decimals = 2
  ) |>
```

```r
  cols_label(
    result = "Game Result",
    Avg_Running_Yards = "Avg Running Yards",
    Avg_Points = "Avg Points"
  )


ggplot(teamSeasonAnalysis, aes(x = offense_ave_points, y = win_pct)) +
  geom_point(size = 3, alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Relationship Between Offensive Scoring and Win Percentage",
    x = "Average Offensive Points",
    y = "Win Percentage"
  ) +
  theme_minimal()

topPassingTeams <- teamSeasonAnalysis |>
  group_by(team) |>
  summarise(
    avg_pass_yards = mean(offense_ave_yards_gained_pass, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(desc(avg_pass_yards)) |>
  slice_head(n = 10)

ggplot(topPassingTeams,
       aes(x = reorder(team, avg_pass_yards),
           y = avg_pass_yards)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Top 10 NFL Teams in Average Passing Yards",
    x = "Team",
    y = "Average Passing Yards"
  ) +
  theme_minimal()

topRunningTeams <- teamSeasonAnalysis |>
  group_by(team) |>
  summarise(
    avg_run_yards = mean(offense_ave_yards_gained_run, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(desc(avg_run_yards)) |>
  slice_head(n = 10)
```

```r
ggplot(topRunningTeams,
       aes(x = reorder(team, avg_run_yards),
           y = avg_run_yards)) +
  geom_col(fill = "darkgreen") +
  coord_flip() +
  labs(
    title = "Top 10 NFL Teams in Average Running Yards",
    x = "Team",
    y = "Average Running Yards"
  ) +
  theme_minimal()

ggplot(winLossTotals,
       aes(x = "", y = n, fill = result)) +
  geom_col(width = 1, color = "white") +
  coord_polar(theta = "y") +
  labs(
    title = "Overall NFL Win vs Loss Distribution (2018-2022)",
    fill = "Game Outcome"
  ) +
  theme_void()
```